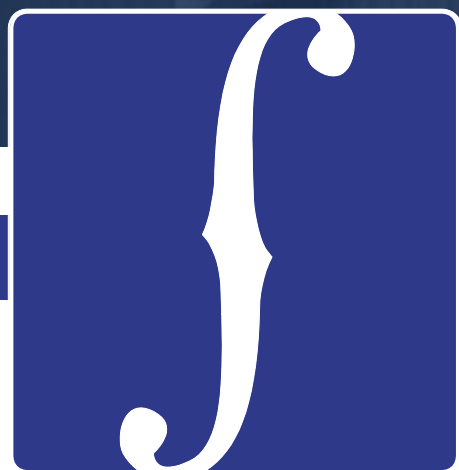




CELLO

LOSS SEVERITY MODEL

October 2015

- Introduction
- Data Description
- Empirical Observations on Loss Severity
- Modeling Framework
- Tracking Results

1

Loss Severity

When a borrower defaults on their loan, Loss Severity expresses the loss as a percentage of the unpaid loan balance:

$$\text{Loss Severity} = \frac{\text{Loss}}{\text{Unpaid Principal Balance}} \times 100\%$$

2

Modeling Loss Severity

- Detailed loss information is available for both Fannie and Freddie since 2000 but in different data formats.
- We decompose Loss Severity into five components and measure/model these components individually.
- We also estimate different submodels for Foreclosure and REO loans as the loss characteristics for these two states are quite different.

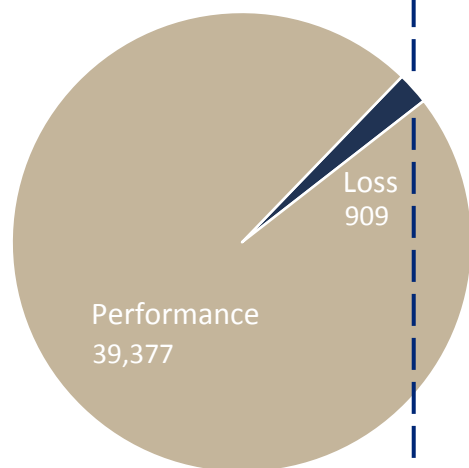
Performance Data:

- Monthly loan performance data is available for Fannie and Freddie at the loan level
- ~ 39 million loans with data going as far back as 2000 (17 million FHL, 22 million FNM)

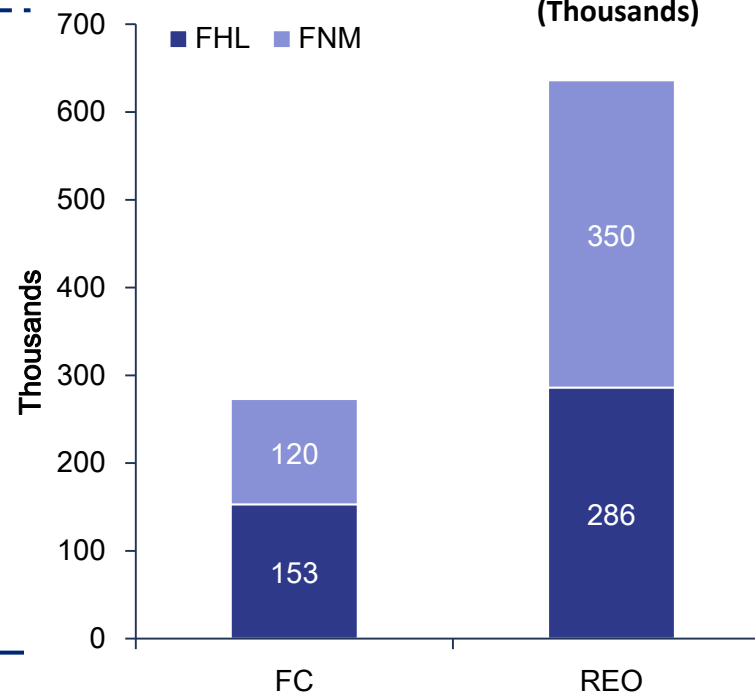
Loss Data:

- Various individual components of loss severity are available for both Fannie and Freddie
- Data is available for Foreclosure and REO sales
- Information on ~ 909,000 loans going as far back as 2000

Relative Share:
Performance vs Loss Data
(Thousands)



Loss Data: FCL vs REO Sales
(Thousands)



11 Components

Current Unpaid Principal Balance	The actual outstanding unpaid principal balance of the mortgage loan.
Net Sale Proceeds	Total cash received from the sale of the property net of any applicable selling expenses
Uncollected Interest	Missed interest payments between the time of the last timely monthly payment and the time the house was liquidated
Foreclosure Costs	Expenses associated with obtaining title to the property from the mortgagor, valuing the property, and maintaining utility services
Property Preservation	Costs for securing and preserving the property, including maintenance and repairs
Asset Recovery Cost	Expenses associated with removing occupants and personal property from an occupied property post foreclosure
Miscellaneous Holding Expenses	Flood, hazard, MI premiums and refund and rental income
Associated Tax	Payment of taxes associated with holding the property
Insurance Payment	Recoveries from mortgage insurance
Repurchase Make Whole	Repurchase of the mortgage loan or loss reimbursement subsequent to property disposition
Other Proceeds	Amount received other than sale proceeds, including redemption proceeds

11 Components

Current Unpaid Principal Balance

Net Sale Proceeds

Uncollected Interest

Foreclosure Costs

Property Preservation

Asset Recovery Cost

Miscellaneous Holding Expenses

Associated Tax

Insurance Payment

Repurchase Make Whole

Other Proceeds

5 Groups

Base Severity = Current Unpaid Principal Balance - Net Sale Proceeds

Uncollected Interest

Liquidation Expenses = Foreclosure Costs + Property Preservation
+ Asset Recovery Cost + Miscellaneous Holding Expenses
+ Associated Tax

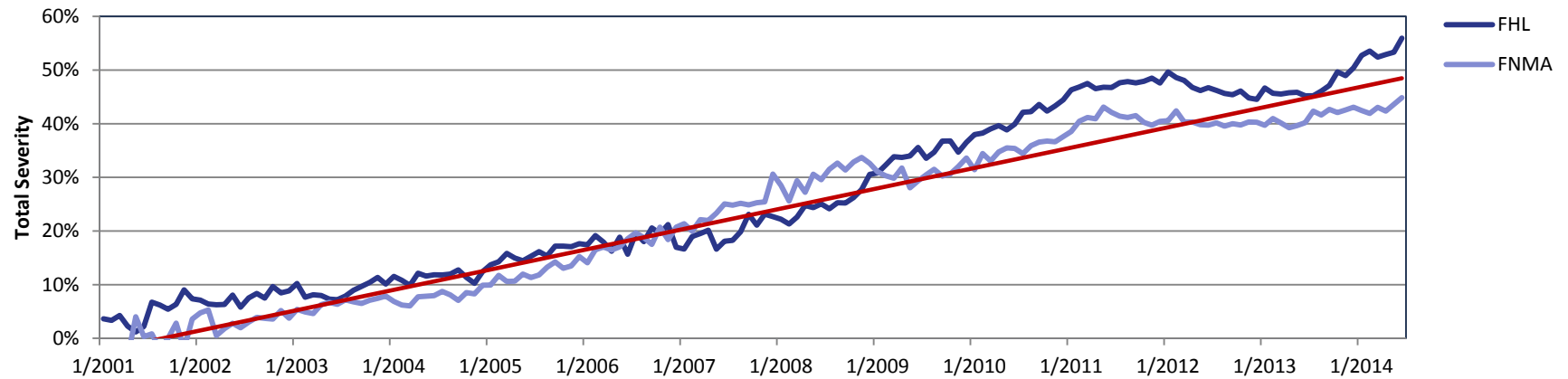
Insurance Payment

Other Foreclosure Proceeds = Repurchase Make Whole + Other Proceeds

- **Total Severity** Historic severity is 36.6%. Increased from 2% in 2001 all the way to 45% in 2014.
- **Base Severity** Averages around 34%. Peaked at 2012, recovered after that. —
- **Uncollected Interest** Significantly increased after 2012 due to an increase of the time from delinquency to liquidation.
- **Liquidation Expenses** All the five subcomponents ramp up after 2005 to double the levels of 2001.
- **Insurance Payment** Noticeable decreases occur after 2007, further increasing loss severity.

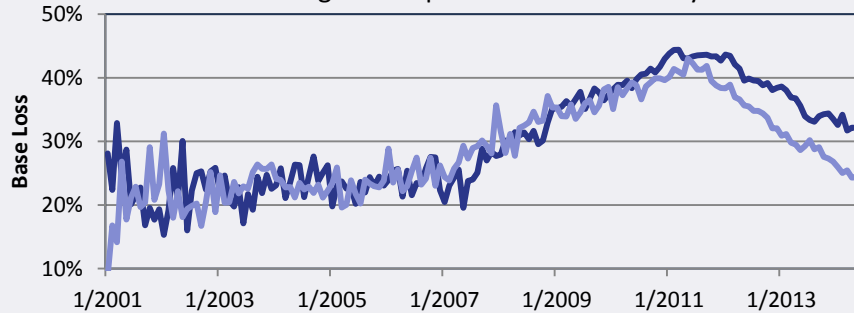
Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	Total
EXPENSES (% UPB)															
Delinquent Interest	8%	9%	9%	9%	9%	10%	9%	8%	8%	8%	9%	10%	12%	15%	10%
Total Liquidation Expenses	6%	6%	7%	7%	9%	10%	9%	7%	6%	6%	7%	8%	11%	14%	8%
Foreclosure	2%	3%	3%	3%	4%	4%	4%	3%	2%	2%	2%	3%	3%	4%	3%
Property Preserve	2%	1%	1%	2%	2%	3%	2%	2%	1%	1%	2%	2%	2%	3%	2%
Asset Recovery	0%	1%	0%	0%	1%	1%	0%	0%	0%	1%	1%	1%	1%	1%	1%
Misc Holding Expenses	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	1%	1%	1%	2%	1%
Associated Tax	1%	1%	1%	1%	2%	2%	2%	2%	2%	2%	2%	2%	3%	4%	2%
Total Cost	113%	115%	115%	116%	118%	119%	118%	115%	113%	114%	116%	118%	123%	129%	118%
PROCEEDS (%UPB)															
Net Sale Proceeds	76%	80%	77%	78%	79%	78%	76%	71%	63%	62%	59%	64%	71%	74%	66%
Insurance Payment	16%	16%	16%	15%	15%	16%	14%	11%	9%	8%	7%	7%	7%	7%	8%
Repurchase Make Whole	15%	12%	13%	10%	8%	5%	3%	4%	8%	9%	8%	5%	3%	1%	6%
Other	4%	4%	4%	5%	5%	5%	5%	2%	1%	1%	1%	1%	2%	2%	2%
Total Proceeds	111%	112%	110%	108%	108%	104%	99%	89%	81%	79%	75%	77%	82%	84%	82%
SEVERITY	2.3%	3.3%	5.9%	7.3%	10.2%	15.2%	19.5%	26.6%	31.9%	34.8%	41.3%	41.0%	40.5%	45.4%	36.6%

Source: Fannie Mae

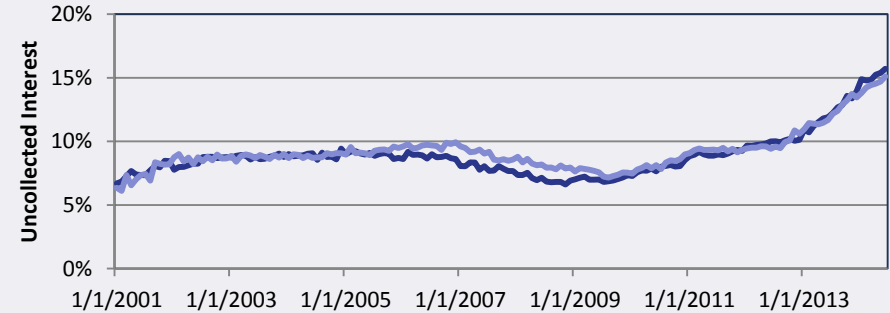


Components

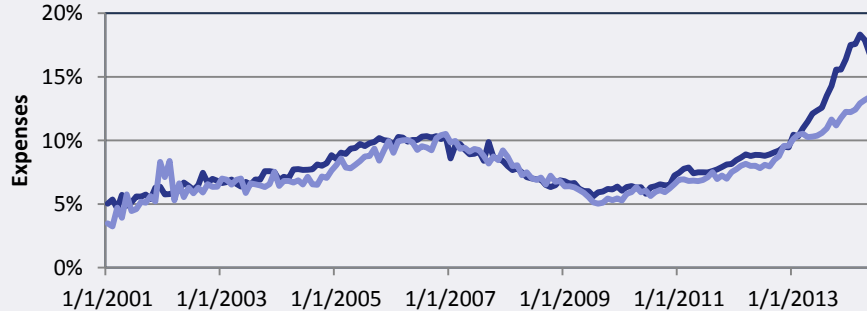
- Base loss is the largest component of Loss Severity



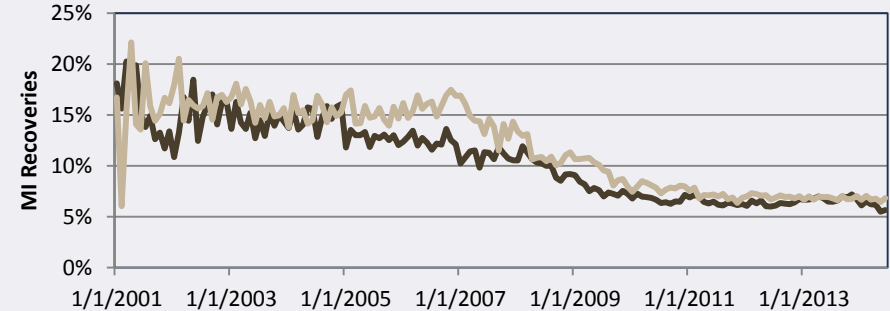
- Stable before 2012 and then doubles



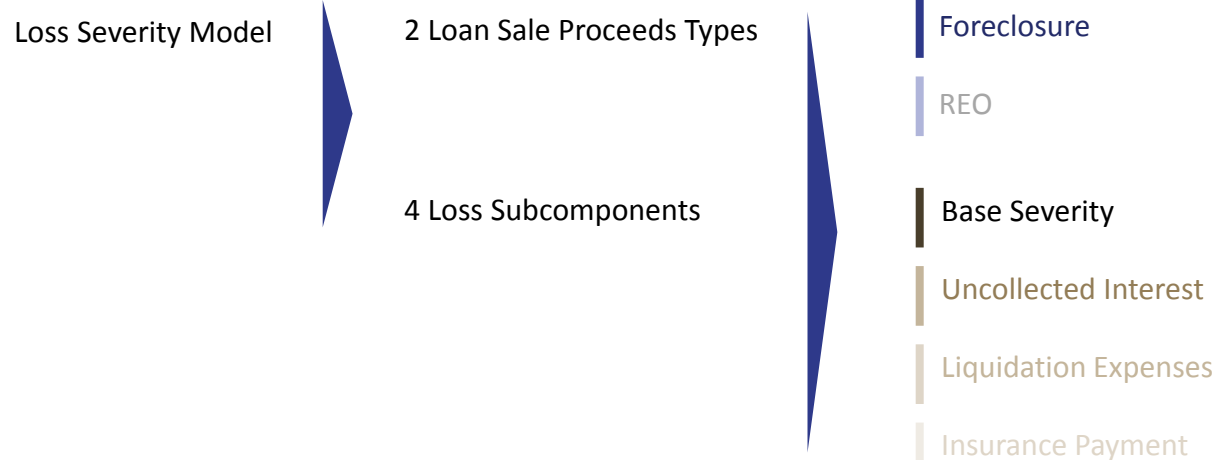
- Overall expenses double after 2012



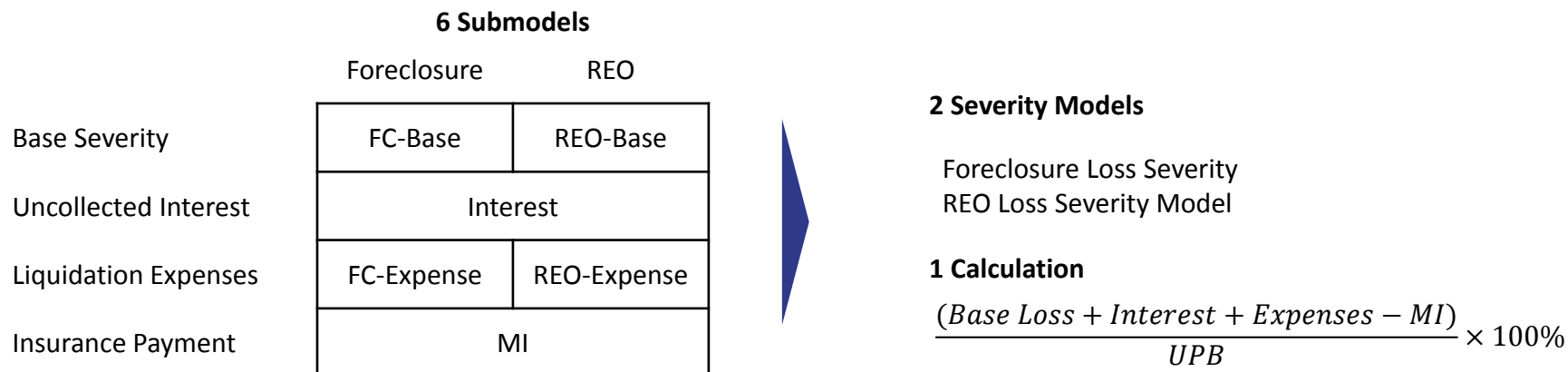
- MI Recoveries decreased from 15% to 7%.



1 Model Decomposition



2 Model Framework



1 Base Severity Calculation

- Base Severity is the largest contributor to Loss Severity, it averages ~35% .
- Base Severity equals the difference between the unpaid loan balance and the sale price of the home

$$\text{Base Loss Amount} = \text{UPB} - \text{Home Sale Price}$$

$$\text{Home Sale Price} = \text{Current Home Value} \times \text{Haircut}$$

Need a regression model to estimate

$$\text{Current Home Value} = \text{Origination Home Value} \times \text{HPI \% Change}$$

2 Factors Affecting Base Severity

► Foreclosure Laws

Loan Size

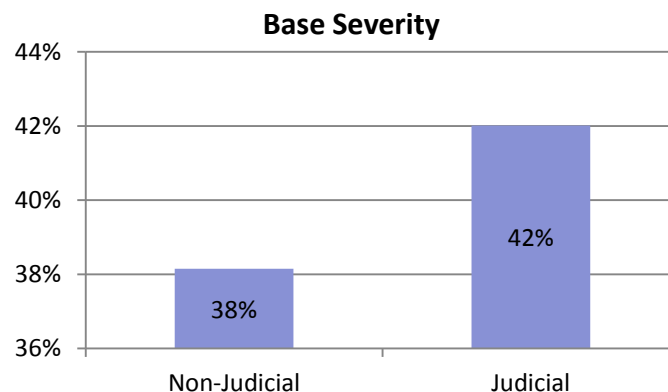
Loan purpose

Occupancy Status

Property Type

Time Since Delinquency

Market Environment



States with Judicial foreclosure laws experience larger differences between HPI-estimated home values and the ultimate sales price.

1 Base Severity Calculation

- Base Severity is the largest contributor to Loss Severity, it averages ~35%.
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$$\text{Base Loss Amount} = \text{UPB} - \text{Home Sale Price}$$

$$\text{Home Sale Price} = \text{Current Home Value} \times \text{Haircut}$$

Need a regression model to estimate

$$\text{Current Home Value} = \text{Origination Home Value} \times \text{HPI \% Change}$$

2 Factors Affecting Base Severity

Judicial

► Loan Size

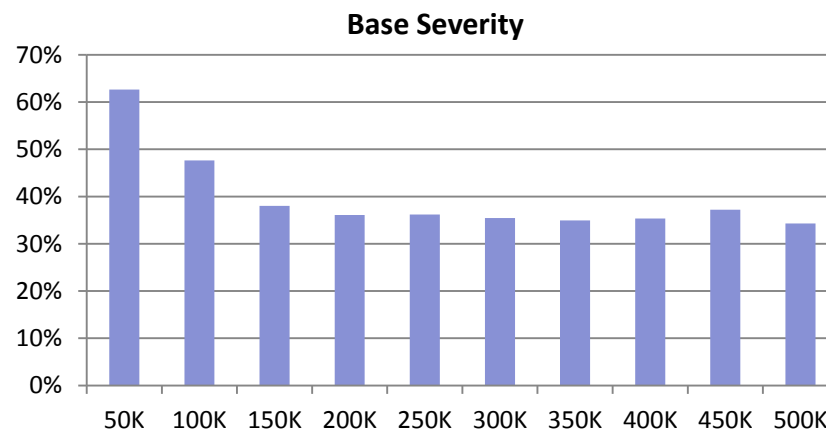
Loan purpose

Occupancy Status

Property Type

Time Since Delinquency

Market Environment



Higher loan sizes are empirically seen to be associated with lower loss severities since the end investor has a greater stake in resolving the situation because of the larger absolute dollar of loss to consider.

1 Base Severity Calculation

- Base Severity is the largest contributor to Loss Severity, it averages ~35%.
- Base Severity equals the difference between the unpaid balance and the sale price of the home.

$$\text{Base Loss Amount} = \text{UPB} - \text{Home Sale Price}$$

$$\text{Home Sale Price} = \text{Current Home Value} \times \text{Haircut}$$

Need a regression model to estimate

$$\text{Current Home Value} = \text{Origination Home Value} \times \text{HPI \% Change}$$

2 Factors Affecting Base Severity

Judicial

Loan Size

► **Loan Purpose**

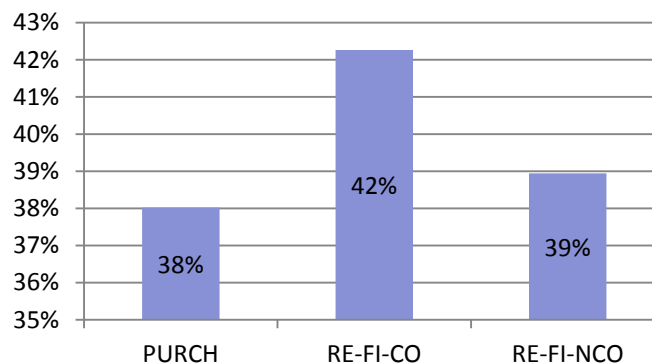
Occupancy Status

Property Type

Time Since Delinquency

Market Environment

Base Severity



Cash-out refinance loans experience somewhat higher severities

1 Base Severity Calculation

- Base Severity is the largest contributor to Loss Severity, it averages ~35%.
- Base Severity equals the difference between the unpaid balance and the sale price of the home.

$$\text{Base Loss Amount} = \text{UPB} - \text{Home Sale Price}$$

$$\text{Home Sale Price} = \text{Current Home Value} \times \text{Haircut}$$

Need a regression model to estimate

$$\text{Current Home Value} = \text{Origination Home Value} \times \text{HPI \% Change}$$

2 Factors Affecting Base Severity

Judicial

Loan Size

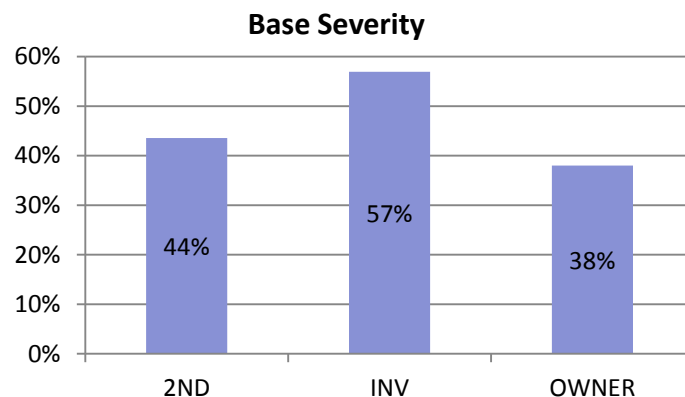
Loan purpose

► **Occupancy Status**

Property Type

Time Since Delinquency

Market Environment



Investor properties are associated with higher loss severities

1 Base Severity Calculation

- Base Severity is the largest contributor to Loss Severity, it averages ~35%
- Base Severity equals the difference between the unpaid loan balance and the sale price of the home.

$$\text{Base Loss Amount} = \text{UPB} - \text{Home Sale Price}$$

$$\text{Home Sale Price} = \text{Current Home Value} \times \text{Haircut}$$

Need a regression model to estimate

$$\text{Current Home Value} = \text{Origination Home Value} \times \text{HPI \% Change}$$

2 Factors Affecting Base Severity

Judicial

Loan Size

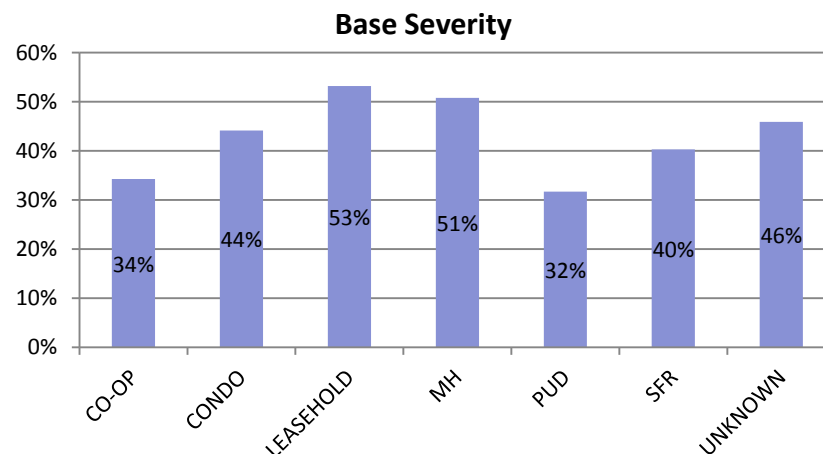
Loan purpose

Occupancy Status

► **Property Type**

Time Since Delinquency

Market Environment



Single-family residences may be more liquid

1 Base Severity Calculation

- Base Severity is the largest contributor to Loss Severity, it averages ~35%.
- Base Severity equals the difference between the unpaid loan balance and the sale price of the home.

$$\text{Base Loss Amount} = \text{UPB} - \text{Home Sale Price}$$

$$\text{Home Sale Price} = \text{Current Home Value} \times \text{Haircut}$$

Need a regression model to estimate

$$\text{Current Home Value} = \text{Origination Home Value} \times \text{HPI \% Change}$$

2 Factors Affecting Base Severity

Judicial

Loan Size

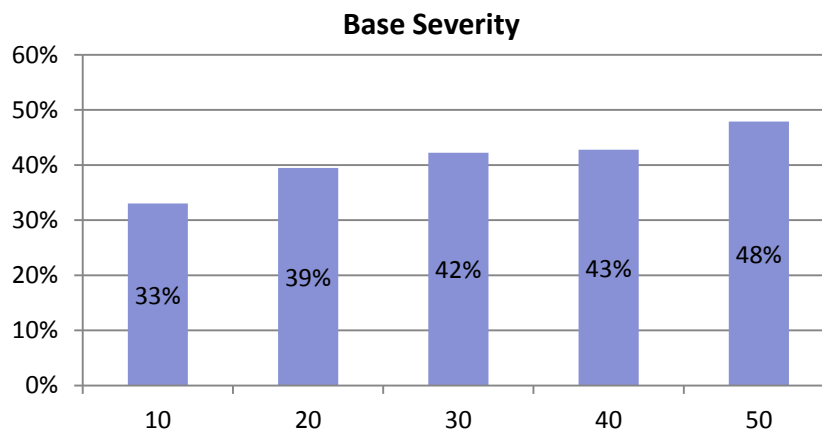
Loan purpose

Occupancy Status

Property Type

► **Time Since Delinquency**

Market Environment



Home associated with longer mortgage delinquency histories may be in worse condition due to lack of maintenance

1 Base Severity Calculation

- Base Severity is the largest contributor to Loss Severity, it averages ~35%
- Base Severity equals the difference between the unpaid balance and the sale price of the home.

$$\text{Base Loss Amount} = \text{UPB} - \text{Home Sale Price}$$

$$\text{Home Sale Price} = \text{Current Home Value} \times \text{Haircut}$$

Need a regression model to estimate

$$\text{Current Home Value} = \text{Origination Home Value} \times \text{HPI \% Change}$$

2 Factors Affecting Base Severity

Judicial

Loan Size

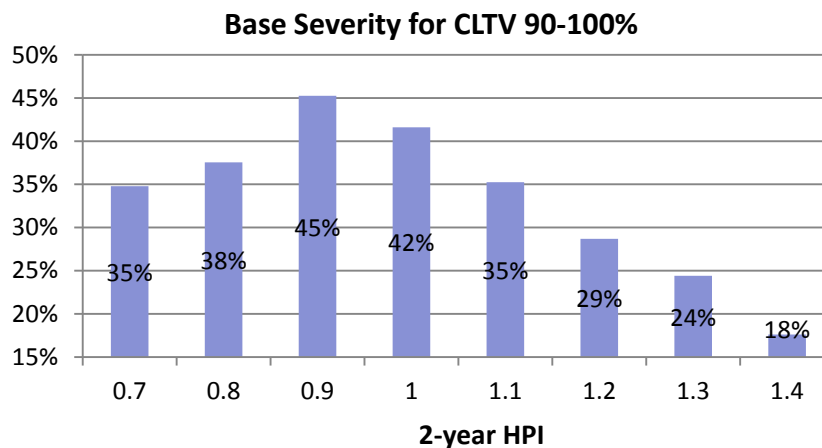
Loan purpose

Occupancy Status

Property Type

Time Since Delinquency

► **Market Environment**

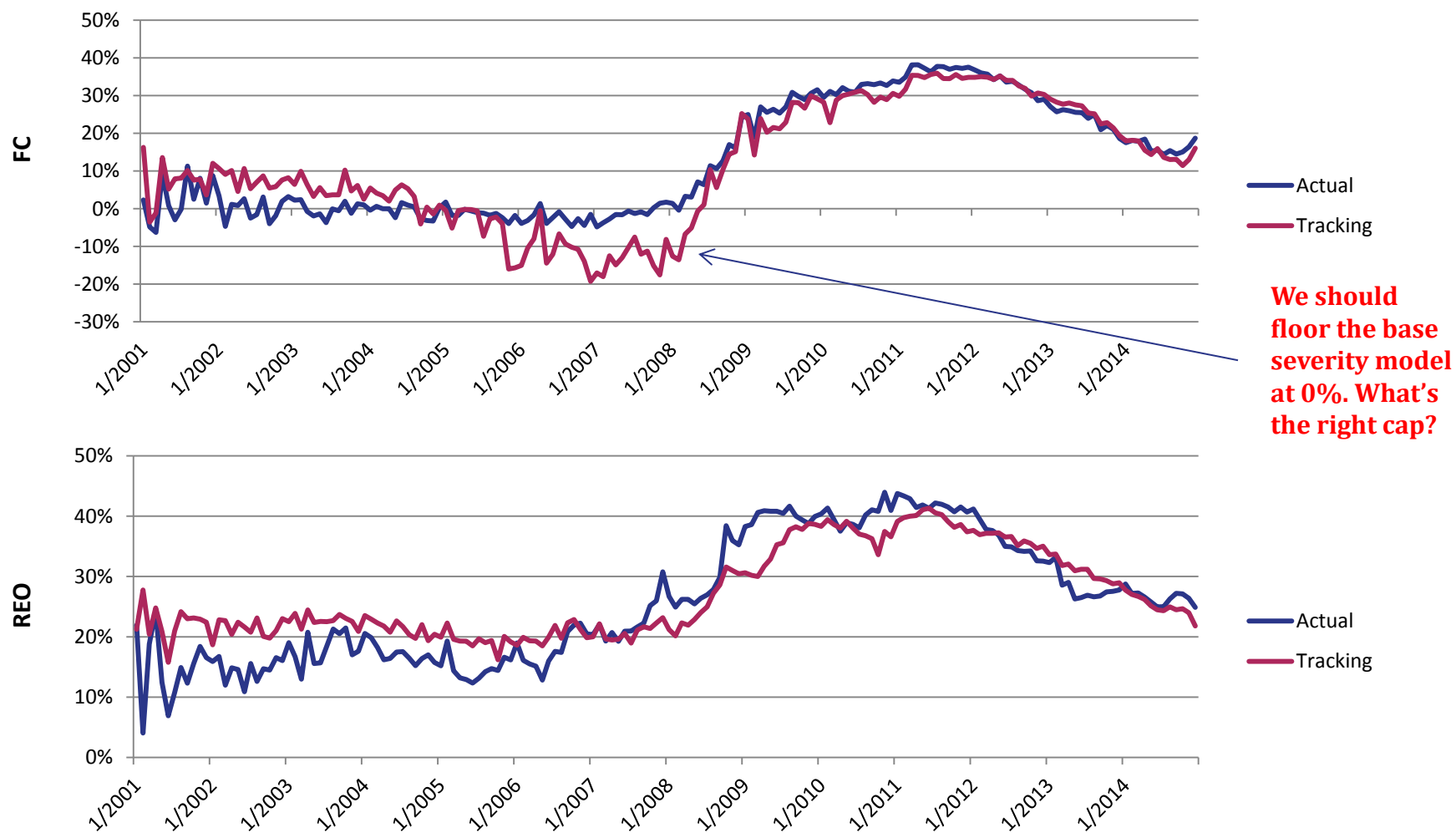


We use a 2-year HPI % change to capture the broader housing market environment. In bad markets it is more difficult to sell a house and the price haircut should be greater.

3

Base Severity Regression Model (based on Fannie Mae data)

- We fit a general linear regression model that incorporates the 7 factors mentioned above for the haircut parameter and then calculate the base severity



1 Liquidation Expenses Calculation

- Liquidation expenses include 5 elements:

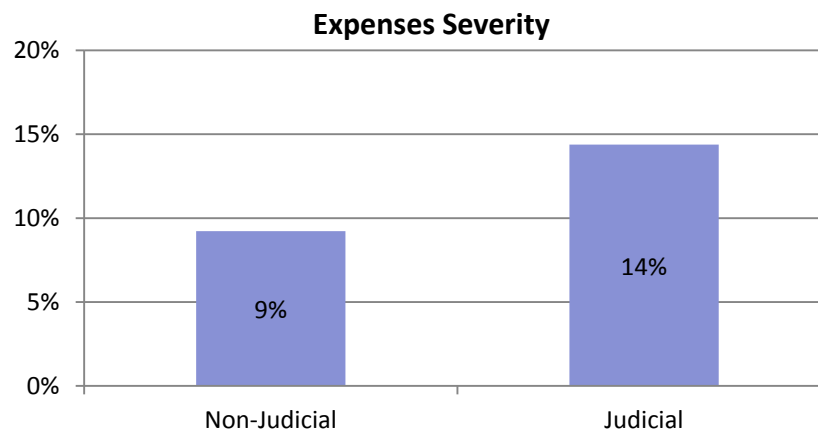
Foreclosure Costs	Expenses associated with obtaining title to property from the mortgagor, valuing the property, and maintaining utility services
Property Preservation	Cost for securing and preserving the property, including maintenance and repairs
Asset Recovery Cost	Expenses associated with removing occupants and personal property from an occupied property post foreclosure
Miscellaneous Holding Expenses	Flood, hazard, MI premiums and refund and rental income
Associated Tax	Payment of taxes associated with holding the property

2 Factors Affecting Magnitude of Expenses

► Foreclosure Laws

Loan Size

Time Since Delinquency



Foreclosure expenses vary by state because of the difference in property laws by state. States with a Judicial process will experience a higher loss severity.

2

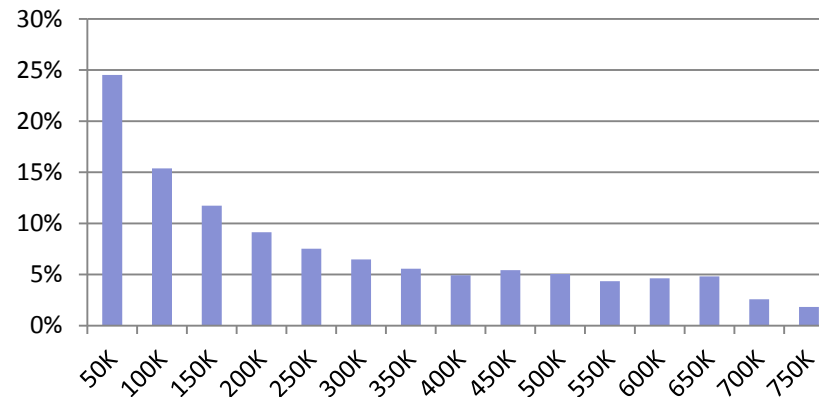
Factors Affecting Severity of Expenses

Judicial

► **Loan Size**

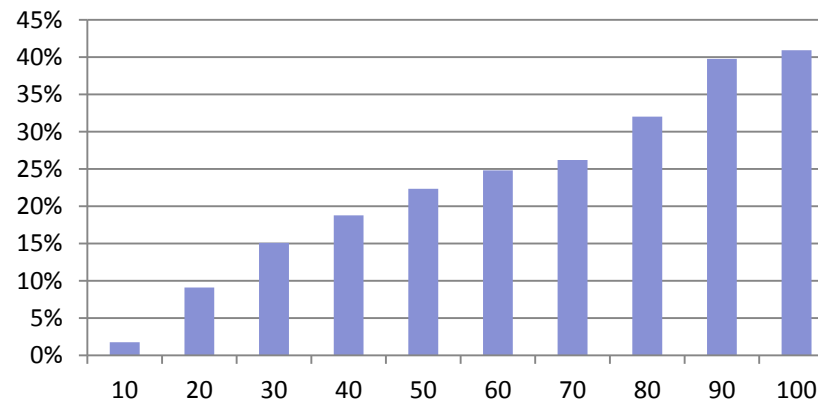
Time Since Delinquency

Expenses Severity



Higher loan sizes are empirically seen to be associated with lower loss severities since the end investor has a greater stake in resolving the situation because of the larger absolute dollar of loss to consider.

Expenses Severity



Holding costs will depend upon time spent in delinquency

Judicial

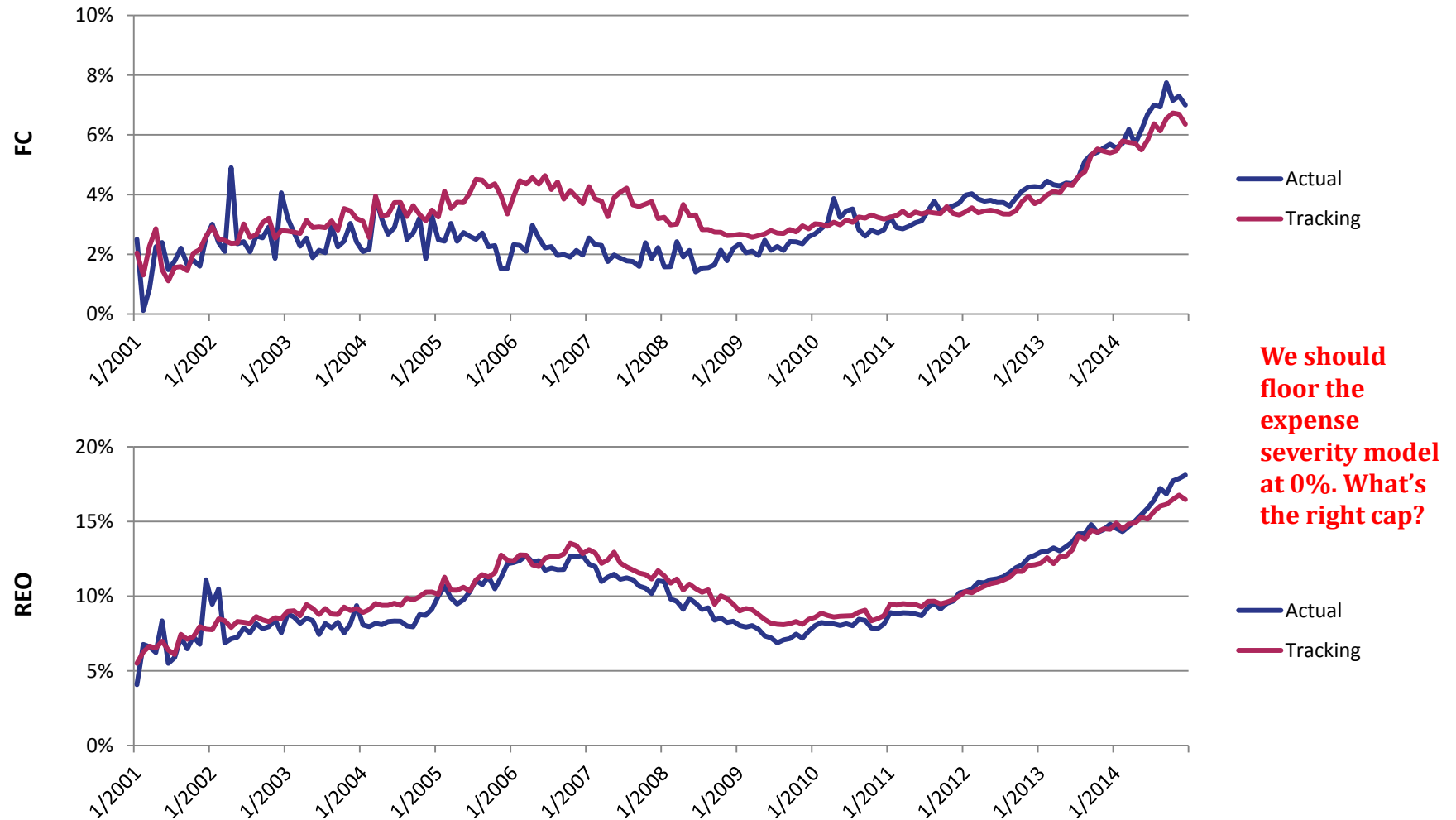
Loan Size

► **Time Since Delinquency**

3

Expenses Regression Model

- We use a regression model to estimate the expenses from the 3 factors mentioned above.



- REO loans have a higher severity from expenses due to a longer resolution period.

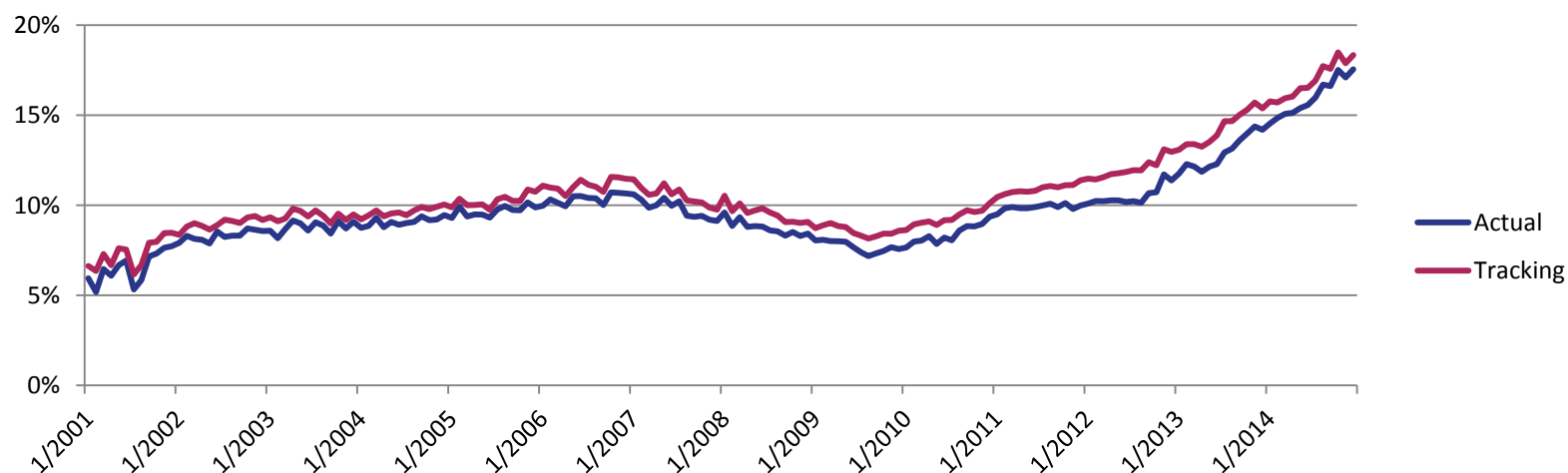
1 Uncollected Interest Calculation

- The amount of Uncollected Interest depends both on the loan coupon and the months of missed payments.
 - Coupon and servicing fee are fixed at origination
 - Months of Lost Interest will be calculated from the roll rate model.

$$\text{Uncollected Interest} = \frac{(\text{Coupon} - \text{Servicing Fee})}{12} \times \text{Months of Lost Interest}$$

2 Model Tracking Results for Uncollected Interest

**Model has
floors and
caps?**



1 Insurance Payment Calculation

- In general, a PMI policy covers only a specified portion of the loss. This percentage is typically between 12% to 35% of the loss claim. Most lenders use the loss coverage scheme developed by Fannie Mae and Freddie Mac. For example, the standard coverage on a 25- to 30-year loan is as follows:

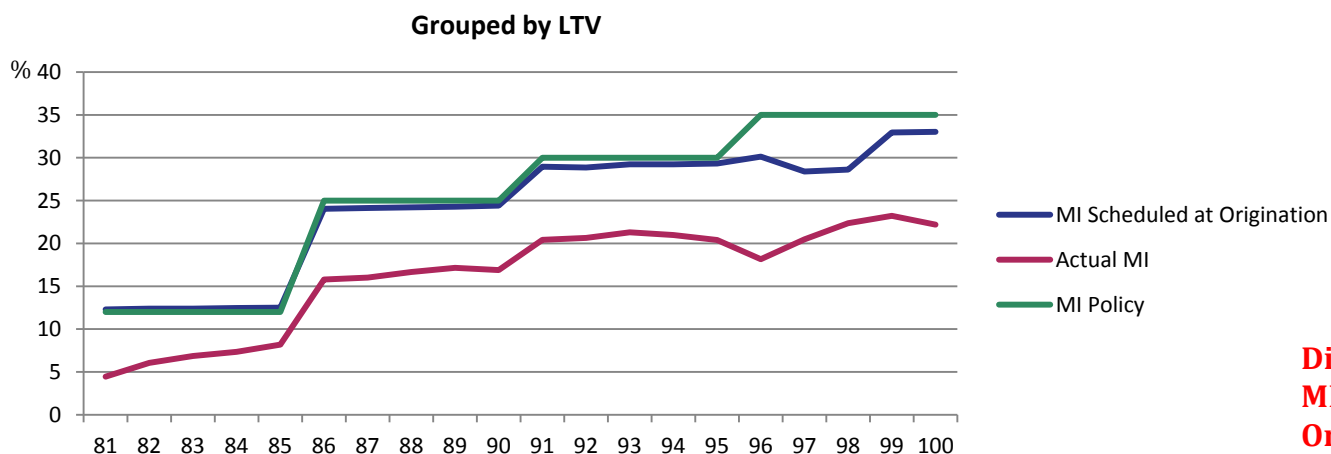
Coverage	LTV RANGE	
35%	>= 95.01	<= 100
30%	90.01	95
25%	80.01	90
12%	80	85

- The coverage percentage is based on **Claim Amount**:

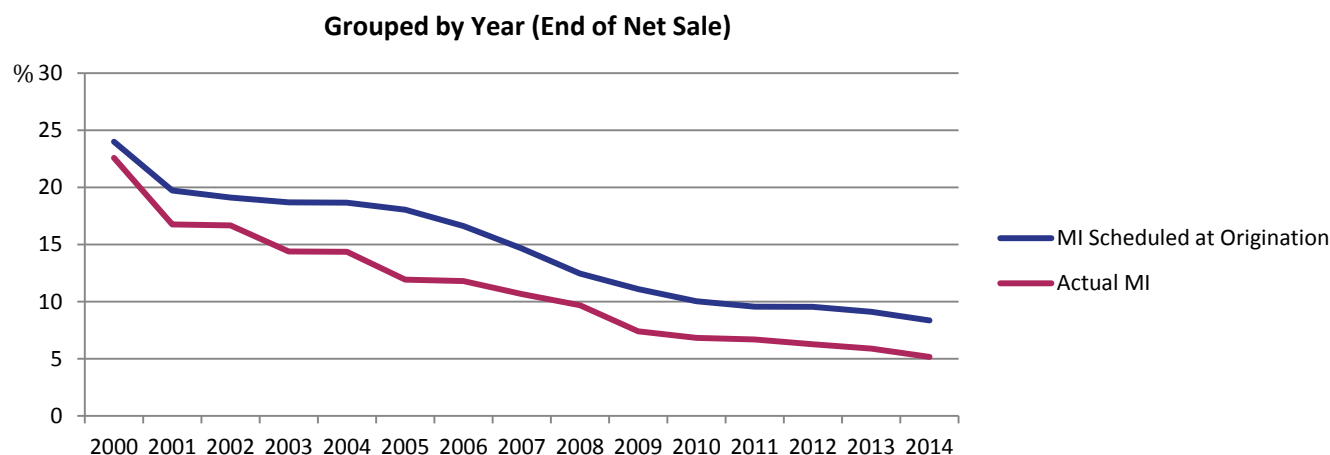
$$\text{Claim Amount} = \text{Unpaid balance} + \text{Uncollected Interest} + \text{Liquidation Expenses}$$

2 Analysis of MI Coverage

- In practice, the actual coverage amount appears to vary significantly from the scheduled MI amount.



**Difference between
MI Scheduled at
Origination and MI
Policy?**



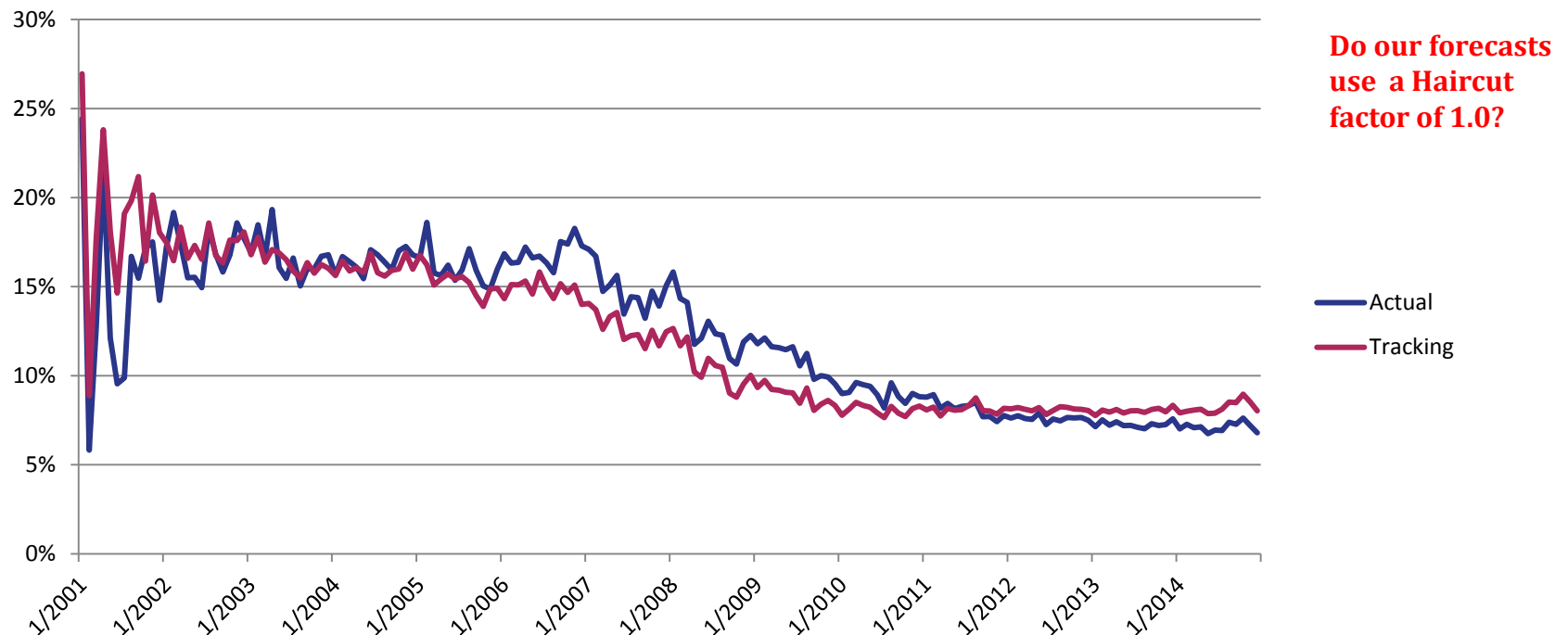
- Comparing scheduled vs actual coverage reveals a 20-30% difference

3

Fitting MI Recoveries

- MI Payment Modeling needs to take into account the fact that there may be discrepancies between the actual and scheduled recoveries.

$$\text{Estimated MI} = \text{Estimated Claim Amount} * \text{Origination MI Level} * \text{MI Haircut Factor}$$



- In historical tracking, we use an MI Haircut Factor equal to 0.8. Reasons for the lower payments include: (1) Deferred payment options for financial distressed MIs, (2) Claim denials, and (3) Claim curtailment because the foreclosure process was longer than it should have been. However, going forward, GSEs are effectively backstopping the MI payments in their CRT transactions.

1

Other Foreclosure Proceeds Definition

- **Fannie Mae**

- **Repurchase Make Whole Proceeds:** Amounts received for the repurchase of the mortgage loan or the subject property or loss reimbursement subsequent to property disposition.
- **Other Foreclosure Proceeds:** Amounts, other than sale proceeds, received by Fannie Mae following foreclosure of a property, including redemption proceeds received from the mortgagor.

- **Freddie Mac**

- Proceeds received based on repurchase/make whole proceeds, non-sale income such as refunds (tax or insurance), hazard insurance, rental receipt, positive escrow and other miscellaneous credits.

- **In other words, there is no clear way to predict this “Other Foreclosure Proceeds”**

2

A Proxy for Other Foreclosure Proceeds

- As there is no clear way to predict this number, we use an approximation based on MI recoveries.



The issue is that MI recoveries are a function of initial LTV. Same for non-MI recoveries?

- These two recoveries proceeds share same downward trend;
- Other Foreclosure Proceeds is much more volatile;
- Other Foreclosure Proceeds is lower than MI recoveries after 2005.

We use 0.75 times MI recoveries as a proxy.

Data Inputs for Model Forecasts

Models	Components	Required Inputs	Sources
Base Severity	UPB	UPB	Schedule
		Origination Home Value	Origination
	Haircut	HPI Changes	HPI Assumption
		Foreclosure Law	Origination
		Loan Size	Origination
		Loan Purpose	Origination
		Occupancy Status	Origination
		Property Type	Origination
		Time Since Delinquency	Roll Rate Model
		Market Environment	HPI Assumption
Uncollected Interest	Coupon	Coupon	Origination
	Time Since Delinquency	Time Since Delinquency	Roll Rate Model
Liquidation Expenses		Foreclosure Law	Origination
		Loan Size	Origination
		Time Since Delinquency	Roll Rate Model
Insurance Payment	Claim Amount	UPB	Scheduled
		Uncollected Interest	Calculated
		Liquidation Expenses	Calculated
	Haircut	Haircut	Assumption

3 Sources for Model Inputs

Loan-level data released by the Agencies

Assumptions

Roll Rate Model

The Loss Severity model needs to be **integrated with the Roll Rate Model**

A Simple Example of a Roll Rate Calculation (Single Loan)

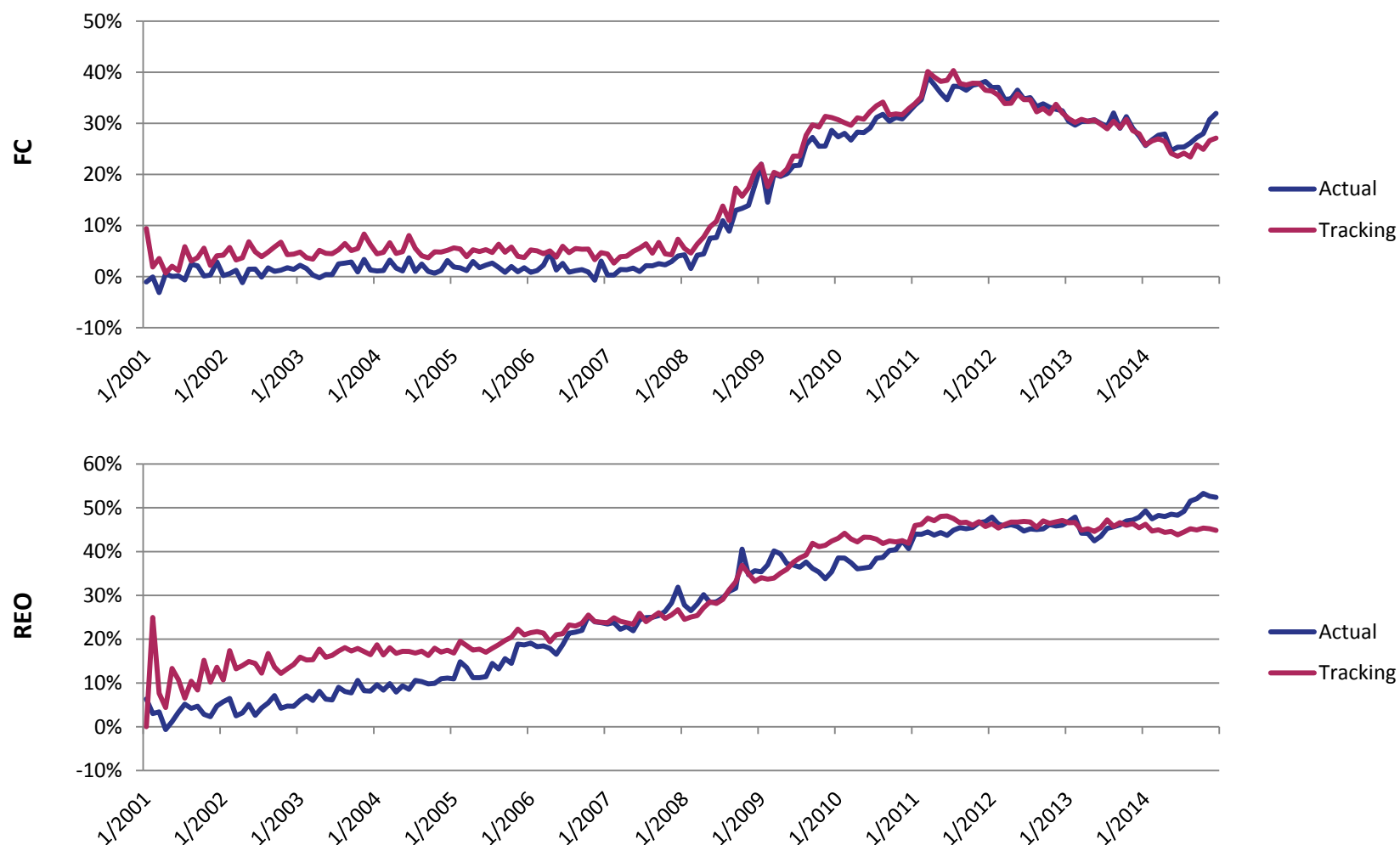
Starting Time (t=0)			Transition Matrix							t=1	
Status	Prob.	Time DQ								Prob.	Time DQ
AC	0	0	AC	C	ED	LD	FC	REO	X	0%	0
C	1	0	95%						5%	91%	0
ED	0	0		91%	7%				2%	7%	1
LD	0	0		49%	36%	13%			2%	0%	0
FC	0	0		11%	14%	67%	6%		1%	0%	0
REO	0	0		3%		2%	92%	2%	1%	0%	0
X	0	0						89%	11%	0%	0
									100%	2%	0

t=2			t=3	
Status	Prob.	Time DQ	Time DQ	
AC	0%	0.00	0.00	
C	86%	0.00	0.00	
ED	9%	1.28	1.47	
LD	1%	2.00	2.53	
FC	0%	0.00	3.00	
REO	0%	0.00	3.00	
X	4%	0.00	0.00	

Input for FC Loss Severity Model	
Input for REO Loss Severity Model	

- After obtaining all five components, we calculate total severity using the following equation:

$$\text{Total Severity} = \frac{(\text{Base Loss} + \text{Interest} + \text{Expenses} - \text{MI} - \text{Other Foreclosure Proceeds})}{\text{UPB}} \times 100\%$$



- Out-of-sample Tracking:**

(Regression on Fannie Mae data and tracking on Freddie Mac data)

